

NUSRAT JAHAN BABLY

Computer Science Graduate | Undergraduate Teaching Assistant | Research & Technology Enthusiast

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nusrat-jahan-bably

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Aspiring Software Engineer with strong interests in full-stack development, AI integration, and research-oriented problem solving. Passionate about building innovative software solutions that create real-world impact while continuously learning, adapting, and contributing in collaborative environments.

WORK EXPERIENCE

Grader & Undergraduate Teaching Assistant

Oct 2024 – Feb 2026

United International University (UIU)

Department of CSE & Data Science

UGA

- Assisted undergraduate students in **Object Oriented Programming** and **Data Structures & Algorithms** labs
- Provided debugging support and academic guidance during lab sessions and helped students strengthen problem-solving and programming fundamentals.
- Evaluated assignments, coding tasks, and class assessments with consistency and fairness.
- Collaborated with faculty members in preparing lab materials and ensuring smooth execution of lab activities.

- Evaluated assignments and class tests with accuracy and consistency, ensuring fair grading aligned with course objectives. Grader
- Provided constructive feedback to help students improve understanding of core **Electrical Circuit** concepts and theoretical foundations.

TECHNICAL PROJECTS

BiblioTheca: An Interactive AI-Powered Gamified Library Management System

Tech: React, JavaScript, Spring Boot, Java, H2 Database, REST API, CSS

GitHub

- Developed a full-stack gamified library management system using React and Spring Boot
- Implemented book management, authentication, routing, and REST API features
- Designed an AI-assisted chatbot module and interactive dungeon mini-game system
- Built responsive dark-mode UI with modern user-centric design

Landlytics: Intelligent Land Analytics Platform

GitHub

Tech: Laravel, PHP, MySQL, Blade, Tailwind CSS, JavaScript, REST APIs

- Developed a Laravel-based platform for land management, agricultural analysis, weather monitoring and crop recommendation.
- Built modules to support data-driven land decision-making and agricultural planning.

Medica DB: A Full-Stack Medical Workflow Management System

GitHub

Tech: React, Node.js, Express, REST APIs, Socket.io, SQL, HTML5 Canvas, JavaScript

- Enabled doctors to manage patients, process medical images, collaborate through professional forums, and monitor activities in real time within a responsive interface.

Fuel Theft & Consumption Monitoring System

Arduino, C/C++, Ultrasonic Sensor, Flow Meter, Speed Sensor, Bluetooth Module (HC-05), I2C LCD

ProjectLink

- Built an IoT system using Arduino to detect unauthorized fuel extraction and track consumption.
- Integrated ultrasonic and flow sensors to monitor fuel levels and trigger instant theft alerts.
- Interfaced a speed sensor to analyze the correlation between vehicle velocity and fuel efficiency.
- Implemented wireless data transmission via Bluetooth and displayed real-time metrics on an I2C LCD

RESEARCH EXPERIENCE

Multimodal Speech Reconstruction for Bengali Stroke Patients

[Ongoing] | Poster

Combining Lip Movements and Audio Signals | Final Year Design Project (Thesis)

- Developing a deep learning-based system to reconstruct fluent Bengali speech from lip movements and degraded audio signals of stroke survivors; achieved **Champion** position in the FYDP poster competition.

ShunoBondhu: A Voice Assistive App for Pedal Rickshaw Pullers

[Ongoing]

Human-Centered Computing Solution Research

- Designing a lightweight voice-assisted mobile application that enables non-literate users to interact through simple voice commands for accessing local services, communication support, and everyday assistance.

SmartBoardVision: An Intelligent Classroom Visibility Enhancement System Using Deep Learning

[Ongoing]

Deep Learning and Digital Image Processing Research

- Building an intelligent classroom text visibility pipeline using YOLOv8, EAST, and CRAFT for detection alongside CLAHE and SRGAN for image enhancement.

EDUCATIONAL QUALIFICATIONS

B.Sc. in Computer Science & Engineering (BSCSE) | Graduation year: 20 Feb, 2026

CGPA: 3.90 / 4.00

United International University (UIU), Dhaka, Bangladesh

Higher Secondary Certificate (HSC)

GPA: 4.83

Ispahani Cantonment Public School and College, Cumilla Cantonment

SSC | JSC

GPA: 5 | GPA: 5

SKILLS

Programming Languages: Python, Java, PHP, JavaScript, SQL, C/C++, HTML, CSS.

Frameworks & Tools: Spring Boot, Laravel, Git/GitHub, Maven, MySQL, H2, XAMPP, LaTeX, Jira, Canva, Microsoft Office.

AI & Data Science: Machine Learning, Deep Learning, NLP, TensorFlow, Pandas, NumPy, Matplotlib, Digital Image Processing.

Research & Development: Literature Review, Experimental Design, Data Preprocessing, Prototype Development, Research Writing.

Core Concepts: OOP, DSA, Software Engineering, DBMS, RDBMS, Artificial Intelligence & Machine Learning, Computer Networks & Security, Full-Stack Development, Gamification Design.

Professional Skills : Communication, Mentoring, Public Speaking, Classroom Facilitation, Team Collaboration, Problem Solving, Time Management.

ACHIEVEMENTS & ADDITIONAL ACTIVITIES

- Champion** – Final Year Design Project at CSE project show.
- Champion** – Intra University Chess Competition.
- Academic Scholarship Recipient** – Earned 100% (6 times), 50% (4 times), and 25% (2 times) tuition waivers based on academic performance at UIU.
- Champion** – 1st Mirror Inter College National Debate Competition | **Participant** – National High School Programming Contest.(ICT Division)

REFERENCES

Dr. Suman Ahmmed

[View LOR]

Head & Associate Professor, Dept. of CSE, UIU

Dr. Mohammad Nurul Huda

[View LOR]

Professor, Dept. of CSE, UIU